Step 6: Prerequisites Quiz

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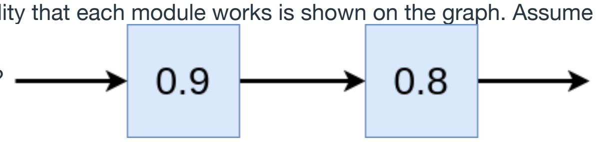
The following quiz will help you judge if you have the appropriate mathematical background for this course. It is okay if you do not answer some questions correctly. More importantly, do you understand the correct answers?

1 Multiple choice 1 point

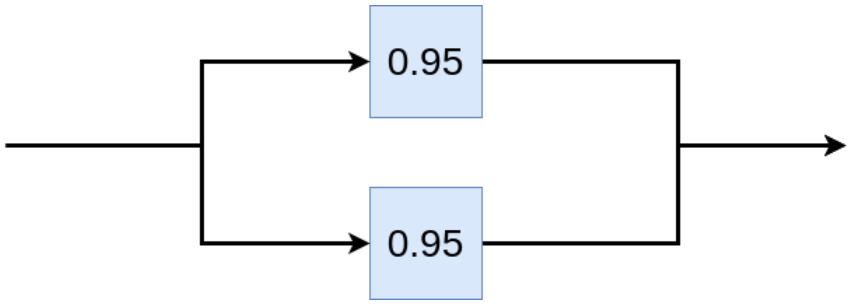
The following two-module system works only if both modules work. The probability that each module works is shown on the graph. Assume that the modules fail independently. What is the probability that the system works?



- 0.81
- 0.9
- 0.8
- 0.72

2 Multiple choice 1 point

This two-module system works if at least one module works. The probability that each module works is shown on the graph. The modules fail independently. What is the probability that the system works?



- ◯ 0.995
- ◯ 0.005
- ◯ 0.9975
- ◯ 0.0025

3 Multiple choice 1 point

A batch of parts contains 100 parts from Supplier A and 200 parts from Supplier B. If 4 parts are selected randomly, without replacement, what is the probability that they are all from Supplier A?

- ◯ 0.9881
- ◯ 0.0119
- ◯ 0.9771
- ◯ 0.0229

4 Multiple choice 1 point

A batch of parts contains 100 parts from Supplier A and 200 parts from Supplier B. If 4 parts are selected randomly, with replacement, what is the probability that they are all from Supplier A?

- 0.9877
- 0.0123
- 0.9771
- 0.0229

5 Multiple choice 1 point

Suppose that the current measurements in a strip of wire are assumed to follow a normal distribution with mean = 10 and variance = 4, what is the probability that a measurement exceeds 13? (Hint. you can use a “z-table” for the standard normal distribution)

- 0.06681
- 0.07982
- 0.05586
- 0.04389

6 Multiple choice 1 point

6. Find the following limit : $lim_{x \to 0} \dfrac{sin(x) + x}{2x^2 + x}$

- 0
- 1/2
- 1/4
- 2

7 Multiple choice 1 point

Differentiate $\ln(x)$ with respect to x,

- $\frac{1}{10x}$
- $\frac{1}{x}$
- 0
- 1

8 Multiple choice 1 point

Integrate a^x with respect to x.

- $a^x + C$
- $a^{x-1} + C$
- $\frac{a^x}{\ln(a)} + C$
- $\frac{a^x}{\ln(x)} + C$

9 Multiple choice 1 point

What is the functional form of the solution to the differential equation : $\frac{dx}{dt} = 5x - 3$

- $x(t) = Ce^{5t} + \frac{3}{5}$
- $x(t) = Ce^{5t}$
- $x(t) = C\ \ln(5t) + \frac{3}{5}$
- $x(t) = C\ \ln(5t)$

10 Multiple choice 1 point

Find the eigenvalues for the matrix :

-2	1
12	-3

- ◯ -1, 6
- ◯ 1, -6
- ◯ 2 , 3
- ◯ -2 , -3

11 Multiple choice 1 point

If matrix A has dimension 3x4 and matrix B has dimension 4x5, what are the dimensions of the matrix $B^T A^T$.

- 3x4
- 4x5
- 3x5
- 5x3

12 Multiple choice 1 point

If a symmetric matrix has the following eigenvalue decomposition, $A = U^T \, Y \, U$, the matrix A^n , can then be computed using :

- $U^{Tn} \, Y^n \, U^n$
- $U^T \, Y^n \, U$
- $U \, Y^n \, U^T$
- All of the above

13 Multiple choice 1 point

Which of the following statements are **TRUE** :

- ◯ The inverse of any given matrix A exists if the matrix is singular.
- ◯ The inverse of any given matrix A exists if the matrix is square symmetric.
- ◯ If A is mxn and $n \leq m$, then the maximum possible rank of the matrix is n.
- ◯ The matrix A has an inverse if the determinant is non-negative.”